MATH 4730 M - Winter 2026 - Project Report

Option 1: Memory Puzzle Experiment

Jason Chen, Anamika , Mihret Gebremedhin

1. Introduction & Hypothesis

1.1 Introduction

Memory is an important thing, rife with studies and interest. Thus, we chose to venture towards a study of memory against a common condition, music. Does musical condition affect the performance of memory? Specifically, soft calming music and loud rock music. We set up these conditions via an experiment of a 24-card memory puzzle, and we measured the moves missed and moves taken in the 24-card memory puzzle to identify memory performance. Using the tools we have studied in the course, we have analyzed the results and identified whether musical conditions have an effect on memory.

1.2 Hypothesis

Our hypothesis for the project is as follows:

Null hypothesis (H_0):
Music condition has no effect on memory performance (mean misses are equal across conditions, and mean moves are equal across conditions)

Given:

- μ_1 is mean misses/moves with no music
- μ_2 is mean misses/moves with soft music
- μ_3 is mean misses/moves with loud music

$$H_0: \mu_1 = \mu_2 = \mu_3$$

Alternative hypothesis (H_1):
At least one music condition leads to different memory performance (mean misses are not equal across conditions, and mean moves are equal across conditions)

Given:

- μ_1 is mean misses/moves with no music
- μ_2 is mean misses/moves with soft music
- μ_3 is mean misses/moves with loud music

$$H_1: \mu_1 \neq \mu_2 \text{ or } \mu_1 \neq \mu_3 \text{ or } \mu_2 \neq \mu_3$$

2. Experimental Design & Methods

2.1 Procedure

Participants were asked about their gender and age, and completed 24-card memory puzzles under three randomized music conditions. Results recorded were number of misses and number of moves.

2.1.1 Participants

There are a total of 30 participants for this experiment, with varying age and gender. The participants were recruited via relation to test takers (ie. Anamika, Jason and Mihret), whether family, friends, schoolmates, or random people of proximity to test-taking sites.

2.1.2 Variables

Independent variable (Treatment):
Music condition: Silent (no music), Soft calming music (50% volume), Loud rock music (100% YouTube volume)
Dependent variable:
Primary: Number of misses, Secondary: Number of moves
Blocking variable: Participant
Additional variable: Age, Gender

2.2 Q-Q Plot & Residuals vs. Fitted Plot

Q-Q Plot and Residual vs Fitted plot were used to assess the normality and constant variance assumptions.

2.3 Statistical Models

Three statistical designs were used in this analysis:

2.3.1 Completely Randomized Design (CRD)

A one-way ANOVA model was used to study the effect of music on memory:

$$Y_{ij} = \mu + \tau_i + \epsilon_{ij}$$

This model tests whether the mean response differs across music conditions.

2.3.2 Randomized Complete Block Design (RCBD)

The RCBD was used to study the effect of music on memory with blocking effects:

$$Y_{ij} = \mu + \tau_i + \beta_j + \epsilon_{ij}$$

where :

- $\tau_i =$ treatment effect (music condition)
- $\beta_j =$ block effect (participant)

2.3.3 Factorial Design

Three-factor factorial model was used to explore other factors such as age and Gender
The factors included in this design:

- Music Condition (3 levels)
- Age (2 levels)
- Gender (2 levels)

$$Y = \mu + Music + Gender + Age + Interaction\ term + \epsilon$$

2.4 Post Hoc Analysis

Post Hoc Analysis using Tukey's HSD was conducted for age, as age was significant. However, no post hoc comparisons were done for music since the ANOVA result was not significant.

3. Data Collection Summary

A total of 30 participants were included in the study. Each participant completed the 24-card memory puzzle under three different music conditions, resulting in the following :

- 30 Participants × 3 Conditions = 90 Observations
- 15 Males & 15 Females
- 19 Young (Below 40) & 11 Old Participants (Above 40)

Misses (Primary endpoint): From the data,

Mean of misses = 12.077

- No Music Misses Mean = 10.233
- Soft Music Misses Mean = 12.733
- Loud Music Misses Mean = 13.267

Moves (Secondary endpoint): From the data,

Mean of moves = 31.155

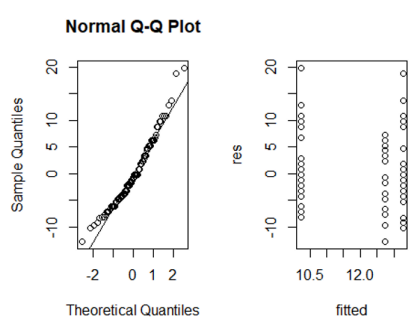
- No Music Moves Mean = 29.067
- Soft Music Moves Mean = 32.133
- Loud Music Moves Mean = 32.267

4. Statistical Analysis & Results

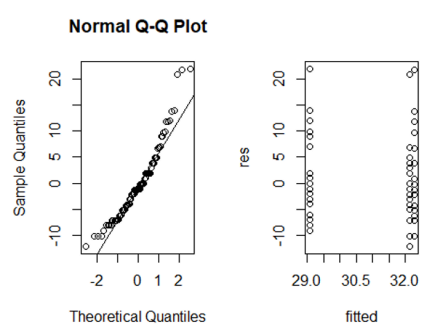
4.1 Q-Q Plot & Residuals vs. Fitted Plot

Q-Q Plot:

Misses (Primary endpoint):



Moves (Secondary endpoint):



- From the plots, points lie close to the reference line in both the Q-Q plots which indicates that the residuals are normally distributed.
- There are very small deviations at the tails for the large values, but normality assumption is satisfied.

Residual vs Fitted Plot:

- Shows that the residuals are randomly scattered around zero. No clear pattern.
- The assumption of constant variance is satisfied.

4.2 Statistical Models

4.2.1 Model 1: Completely Randomized Design (CRD) - Misses (primary endpoint)

```
m1.aov = aov(misses ~ music, data = misses.data)

# Treatment effects
model.tables(m1.aov)

## Tables of effects
##
##  music
## music
##       N       S       L
## -1.8444  0.6556  1.1889

# Hypothesis test
summary(m1.aov)

##             Df Sum Sq Mean Sq F value Pr(>F)
## music        2    157   78.68   1.842  0.165
## Residuals   87   3715   42.70
```

From the ANOVA table,

- F- value = 1.842
- P-value = 0.165

Results: Since the p-value is greater than 0.05, we do not reject the null hypothesis. This shows that there is no statistically significant effect of music on the number of misses.

4.2.2 Model 2: Randomized Block Design (RCBD) - Misses (primary endpoint)

```
# (3) ANOVA table for RCBD
m1 = aov(misses ~ factor(music) + factor(participant), data = RCBD_dataframe)
summary(m1)

##                     Df Sum Sq Mean Sq F value Pr(>F)
## factor(music)        2  157.4   78.68   2.492 0.0916 .
## factor(participant) 29 1883.8   64.96   2.057 0.0098 **
## Residuals           58 1831.3   31.57
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

From the ANOVA Table,

- Treatment (Music Condition): F- value = 2.492, p-value = 0.0916
- Block (Participants): F- value = 2.057, p-value = 0.0098

Results:

The treatment effect (music) is not significant as the p-value is greater than 0.05. However, the p-value is relatively small for participants, indicating that the blocking effect (participants) is statistically significant. This shows that participants differ significantly in the game performance.

4.2.3 Model3: Factorial Design - Misses (primary endpoint)

```
                 Df Sum Sq Mean Sq F value Pr(>F)
music             2    157   78.68   1.856 0.1631
gender            1      1    1.34   0.032 0.8591
age               1    243  242.81   5.728 0.0191 *
music:gender      2     83   41.74   0.985 0.3781
music:age         2     45   22.29   0.526 0.5932
gender:age        1      8    8.14   0.192 0.6625
music:gender:age  2     28   14.03   0.331 0.7193
Residuals        78   3307   42.39
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

From the ANOVA table,

- Music:

F-value = 1.856, p-value= 0.1631

- Gender :

F-value = 0.032,p-value = 0.8591

- Age :

F-value = 5.728, p-value = 0.191

Results:
Only Age is statistically significant with p-value less than 0.05 suggesting that performance differs between younger and older participants. The effect of music and gender is not statistically significant shows that performance does not depend on music and gender. All the interaction terms are not statistically significant indicating the effect of music does not depend on gender or age.

4.2.4 Model 4: Factorial Design + RCBD - Misses (primary endpoint)

```
##                      Df Sum Sq Mean Sq F value  Pr(>F)
## music                 2  157.4   78.68   2.442 0.09688 .
## gender                1    1.3    1.34   0.042 0.83893
## age                   1  242.8  242.81   7.537 0.00828 **
## factor(Participants) 27 1639.6   60.73   1.885 0.02489 *
## music:gender          2   83.5   41.74   1.296 0.28237
## music:age             2   44.6   22.29   0.692 0.50519
## music:gender:age      2   28.1   14.03   0.435 0.64935
## Residuals            52 1675.2   32.22
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

From the ANOVA table,

- Music: F-value = 2.442, p-value= 0.09688
- Gender : F-value = 0.042, p-value = 0.8393
- Age : F-value = 7.537, p-value = 0.00828
- Participants: F-value = 1.885, p-value = 0.02489

Results:
Only Age is statistically significant with p-value less than 0.05 suggesting that performance differs between younger and older participants. The effect of music and gender is not statistically significant shows that performance does not depend on music and gender. Participants (blocking effect) is statistically significant (p-value <0.05) proves that individual performance remains important even after accounting for age and gender. All the interaction terms are not statistically significant.

4.3 Post Hoc Analysis

Since post hoc comparison was only conducted for age, which was significant.

```
TukeyHSD(aov(Misses ~ age, data = data))

##   Tukey multiple comparisons of means
##     95% family-wise confidence level
##
## Fit: aov(formula = Misses ~ age, data = data)
##
## $age
##                diff      lwr      upr     p adj
## Old-Young 3.417863 0.626613 6.209113 0.0169792
```

Results:

- On average, older participants have 3.42 more misses than younger participants.
- This suggests that performance may depend on age with younger participants performing better than older participants.

5. Discussion & Interpretation of Findings

5.1 Key Findings

The results from all models show that music condition does not have a statistically significant effect on memory performance (misses or moves). Similarly, gender was not found to be significant. However, age was statistically significant, indicating that older participants tend to have higher misses and require more moves. This suggests that age plays an important role in memory performance. The significance of the participant effect suggests that individual differences play an important role in memory performance

5.2 Limitations & Possible Improvements

5.2.1 Limitations

- Small sample size
- Device/volume inconsistency. Not all devices used for testing produce the same volume, even at 50%/100% volume on YouTube.
- Samples not randomly selected. Participants mostly relate to the test taker on a personal level (ie, friends and family), which could create a biased sample size.
- Blocking variables are too elementary. The genders of male and female may miss other gendered participants. The age grouping of less than and more than 40 years old puts 40-year-olds and 80-year-olds in the same category, which we would imagine to have drastically different average results.
- We only test two types of music. We have not identified whether music affects memory for all music genres, and decided it doesn't based on two.

5.2.2 Possible Improvements

- Larger sample size
- Better control of the environment and volumes of devices.
- More blocking effects. Perhaps how much sleep a participant gets or if they consider themselves stressed.
- The samples should be randomly selected. No relation to test takers.
- More stratified blocking variables, maybe continuous over discrete for the age group. Perhaps we ask the participants what gender they are instead of whether they are male or female.
- More music types

Appendix

Model 1 : Completely Randomized Design (CRD) - Moves (secondary endpoint)

```
m2.aov = aov(moves ~ music, data = moves.data)
model.tables(m2.aov)

## Tables of effects
##
##  music
## music
##       N       S       L
## -2.0889  0.9778  1.1111

summary(m2.aov)

##             Df Sum Sq Mean Sq F value Pr(>F)
## music        2    197   98.31   1.896  0.156
## Residuals   87   4511   51.85
```

From the ANOVA Table,

- F-value = 1.896
- P-value = 0.156

Results: Again, the p-value is greater than 0.05, we do not reject the null hypothesis.
Again, this shows that there is no statistically significant effect of background music on the number of moves.

Model 2: Randomized Block Design (RCBD) - Moves (secondary endpoint)

```
summary(m1_moves)

##                     Df Sum Sq Mean Sq F value Pr(>F)
## factor(music)        2  196.6   98.31   2.462 0.0941 .
## factor(participant) 29 2195.2   75.70   1.896 0.0193 *
## Residuals           58 2316.0   39.93
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

From the ANOVA Table,

- Treatment (Music Condition) : F- value = 2.462, p-value = 0.0941
- Block (Participants) : F- value = 1.896, p-value = 0.0193

Results:
The treatment effect (music) is not significant as the p-value is greater than 0.05. However, the p-value is relatively small for participants indicating that blocking effect (participants) is statistically significant. This shows that participants differ significantly in the game performance.

```
#
==============================================================================
==============================
# 1. COMPLETELY RANDOMIZED DESIGN: using One way Anova model to test the
effect of background music on memory        performance using a 24 card
memory Puzzel Game using Misses as a primary endpoint(response)
#
==============================================================================
==============================

# Clear workspace
rm(list=ls())
```

```r
# Load needed packages
library(readxl)

## Warning: package 'readxl' was built under R version 4.4.3

library(Hmisc)

## Warning: package 'Hmisc' was built under R version 4.4.3

##
## Attaching package: 'Hmisc'

## The following objects are masked from 'package:base':
##
##     format.pval, units

# Read Excel file
data <- read_excel("C:/Users/mihre/Downloads/Math 4730 Data Collected
Combined.xlsx")

# Check data
print(data)

## # A tibble: 90 × 7
##    Gender Age   Order Misses Moves `Time(s)` Participants
##    <chr>  <chr> <chr>  <dbl> <dbl>     <dbl>        <dbl>
##  1 F      Y     N          4    21        76            1
##  2 F      Y     S         18    36       110            1
##  3 F      Y     L         10    31        87            1
##  4 F      Y     N         10    29        93            2
##  5 F      Y     L         11    30        92            2
##  6 F      Y     S         16    34       120            2
##  7 F      Y     S         16    36       115            3
##  8 F      Y     N         23    43        80            3
##  9 F      Y     L         13    31        61            3
## 10 M      Y     S          9    28        66            4
## # ℹ 80 more rows

names(data)

## [1] "Gender"       "Age"          "Order"        "Misses"       "Moves"
## [6] "Time(s)"      "Participants"

music = factor(data$Order, levels = c("N","S","L"))
misses = data$Misses

# Create dataset
misses.data = data.frame(music = music, misses = misses)

# Display dataset
print(misses.data)
```

```
##    music misses
## 1      N      4
## 2      S     18
## 3      L     10
## 4      N     10
## 5      L     11
## 6      S     16
## 7      S     16
## 8      N     23
## 9      L     13
## 10     S      9
## 11     L      9
## 12     N     10
## 13     L     15
## 14     S     11
## 15     N      3
## 16     L      8
## 17     N      3
## 18     S     15
## 19     N      3
## 20     L     23
## 21     S      0
## 22     L     12
## 23     S     12
## 24     N     19
## 25     S      6
## 26     N      2
## 27     L     11
## 28     N      4
## 29     S      8
## 30     L      3
## 31     N     10
## 32     S     19
## 33     L      9
## 34     N     21
## 35     S     19
## 36     L      4
## 37     N      4
## 38     S      9
## 39     L     11
## 40     N      4
## 41     S     11
## 42     L      9
## 43     N     11
## 44     S     12
## 45     L     18
## 46     N     17
## 47     S      3
## 48     L     27
## 49     N      4
```

```
## 50     S      5
## 51     L      8
## 52     N     11
## 53     S     11
## 54     L     13
## 55     N      7
## 56     S     19
## 57     L      5
## 58     N     20
## 59     S     18
## 60     L     24
## 61     N     13
## 62     S      8
## 63     L     11
## 64     N      6
## 65     S     19
## 66     L     13
## 67     N     21
## 68     S     20
## 69     L     14
## 70     N     30
## 71     S     17
## 72     L     32
## 73     N      8
## 74     S      9
## 75     L     18
## 76     N      9
## 77     S     15
## 78     L     22
## 79     N      2
## 80     S      8
## 81     L      8
## 82     N     12
## 83     S     15
## 84     L     13
## 85     N      9
## 86     S     18
## 87     L     15
## 88     N      7
## 89     S     16
## 90     L      9
```

```r
# Summary by the background music type
summary(misses ~ music, data = misses.data)
```

```
## misses      N= 90
##
## +-------+-+--+--------+
## |       | | N|  misses|
## +-------+-+--+--------+
```

```
## |  music|N|30|10.23333|
## |       |S|30|12.73333|
## |       |L|30|13.26667|
## +-------+-+--+--------+
## |Overall| |90|12.07778|
## +-------+-+--+--------+
```

```r
# (1) Effects Model (ANOVA)
# y_ij = μ + τ_i + ε_ij

m1.aov = aov(misses ~ music, data = misses.data)

# Treatment effects
model.tables(m1.aov)
```

```
## Tables of effects
##
##  music
## music
##       N       S       L
## -1.8444  0.6556  1.1889
```

```r
# Hypothesis test
summary(m1.aov)
```

```
##             Df Sum Sq Mean Sq F value Pr(>F)
## music        2    157   78.68   1.842  0.165
## Residuals   87   3715   42.70
```

```r
# (2) Means Model (Linear Model)
# The means model: y_ij = μ_i + ε_ij

m1.lm = lm(misses ~ music, data = misses.data)

summary(m1.lm)
```

```
##
## Call:
## lm(formula = misses ~ music, data = misses.data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -12.7333  -4.6167  -0.9833   4.0167  19.7667
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)   10.233      1.193   8.577 3.29e-13 ***
## musicS         2.500      1.687   1.482   0.1420
## musicL         3.033      1.687   1.798   0.0757 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
##
## Residual standard error: 6.535 on 87 degrees of freedom
## Multiple R-squared:  0.04063,    Adjusted R-squared:  0.01858
## F-statistic: 1.842 on 2 and 87 DF,  p-value: 0.1646
```

```r
# ANOVA table
anova(m1.lm)
```

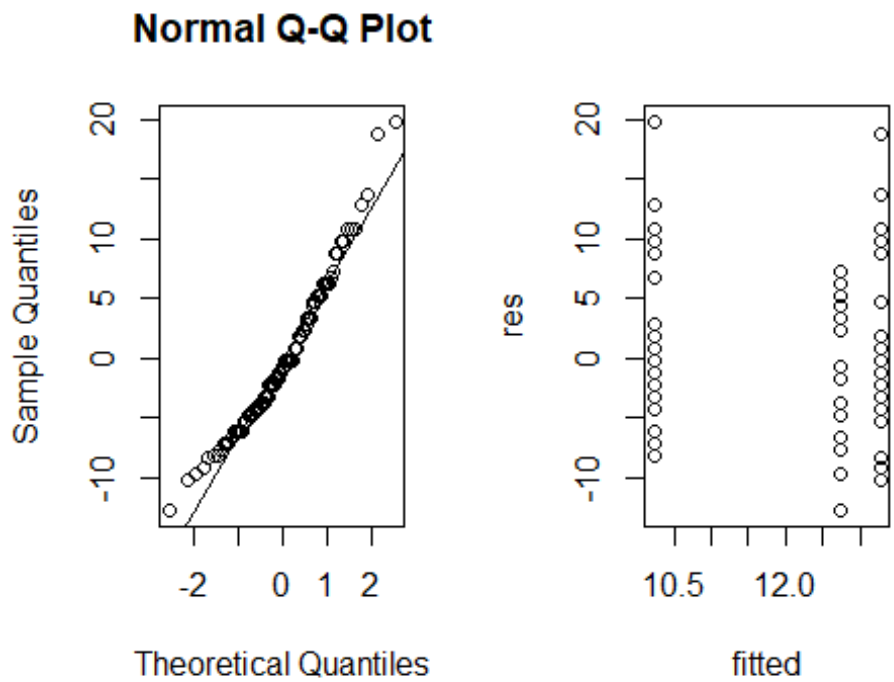
```
## Analysis of Variance Table
##
## Response: misses
##           Df Sum Sq Mean Sq F value Pr(>F)
## music      2  157.4  78.678  1.8425 0.1646
## Residuals 87 3715.1  42.702
```

```r
# To test the effect of the treatment
# The ANOVA table helps test the hypothesis:
# Null Hypothesis (H0): τ_1 = τ_2 = τ_3 = τ_4 = 0  (No effect of background
music on memory performance)
# Alternative Hypothesis (H1): At least one τ_i ≠ 0 (background music has an
effect on memory performance)

# Extract ANOVA table
anova_table <- summary(m1.aov)[[1]]

F0 <- anova_table["music", "F value"]
pvalue <- anova_table["music", "Pr(>F)"]

alpha <- 0.05

# Decision
cat("Decision:", ifelse(pvalue < alpha, "Reject H0", "Do not reject H0"),
"\n")
```

```
## Decision: Do not reject H0
```

```r
# Model Adequacy Checking

res = m1.aov$residuals
fitted = m1.aov$fitted.values

par(mfrow=c(1,2))

# Normality check
qqnorm(res)
qqline(res)

# in the qq plot we can see that the points roughly follow the straight line
```

```r
# Residuals vs fitted
plot(fitted, res)
```




```r
# in the fitted vs residual plot, points are randomly scattred, so all our
assumptions are satisified


#
==============================================================================
==============================
# 2. COMPLETELY RANDOMIZED DESIGN: using One way Anova model to test the
effect of background music on memory        performance using a 24 card
memory Puzzel Game using Moves as a secondary endpoint(response)
#
==============================================================================
==============================

moves = data$Moves
moves.data = data.frame(music = music, moves = moves)

summary(moves ~ music, data = moves.data)

## moves      N= 90
##
## +-------+-+--+--------+
## |       | | N|   moves|
```

```
## +-------+-+--+--------+
## |  music|N|30|29.06667|
## |       |S|30|32.13333|
## |       |L|30|32.26667|
## +-------+-+--+--------+
## |Overall| |90|31.15556|
## +-------+-+--+--------+
```

```r
m2.aov = aov(moves ~ music, data = moves.data)
model.tables(m2.aov)
```

```
## Tables of effects
##
##  music
## music
##       N       S       L
## -2.0889  0.9778  1.1111
```

```r
summary(m2.aov)
```

```
##             Df Sum Sq Mean Sq F value Pr(>F)
## music        2    197   98.31   1.896  0.156
## Residuals   87   4511   51.85
```

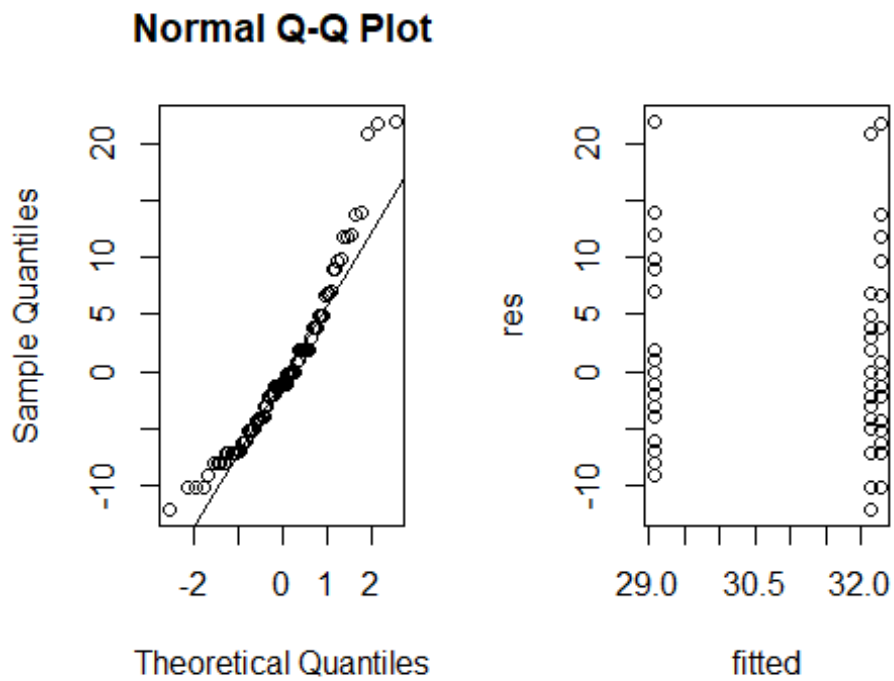
```r
m2.lm = lm(moves ~ music, data = moves.data)
summary(m2.lm)
```

```
##
## Call:
## lm(formula = moves ~ music, data = moves.data)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -12.133  -5.233  -1.100   3.517  21.933
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)   29.067      1.315  22.109   <2e-16 ***
## musicS         3.067      1.859   1.649   0.1027
## musicL         3.200      1.859   1.721   0.0888 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 7.201 on 87 degrees of freedom
## Multiple R-squared:  0.04177,    Adjusted R-squared:  0.01974
## F-statistic: 1.896 on 2 and 87 DF,  p-value: 0.1563
```

```r
anova(m2.lm)
```

```
## Analysis of Variance Table
##
## Response: moves
```

```
##           Df Sum Sq Mean Sq F value Pr(>F)
## music      2  196.6  98.311   1.896 0.1563
## Residuals 87 4511.2  51.853

# To test the effect of the treatment
# The ANOVA table helps test the hypothesis:
# Null Hypothesis (H0): τ_1 = τ_2 = τ_3 = τ_4 = 0  (No effect of background
music on memory performance)
# Alternative Hypothesis (H1): At least one τ_i ≠ 0 (background music has an
effect on memory performance)

# Extract ANOVA table
anova_table <- summary(m2.aov)[[1]]

F1 <- anova_table["music", "F value"]
pvalue1 <- anova_table["music", "Pr(>F)"]

alpha <- 0.05

# Decision
cat("Decision:", ifelse(pvalue < alpha, "Reject H0", "Do not reject H0"),
"\n")

## Decision: Do not reject H0

# Model Adequacy Checking for moves

res = m2.aov$residuals
fitted = m2.aov$fitted.values

par(mfrow=c(1,2))

# Normality check
qqnorm(res)
qqline(res)

# in the qq plot we can see that the points roughly follow the straight line

# Residuals vs fitted
plot(fitted, res)
```




```
# in the fitted vs residual plot, points are randomly scattred, so all our
assumptions are satisified


# Estimating sample size:
# MSE from ANOVA is used as an estimate for variance.
# Effect size: f = d / sqrt(MSE)

library(pwr)

## Warning: package 'pwr' was built under R version 4.4.3

# Number of groups
a <- 3

# Extracting variance from ANOVA (Misses model)
anova_table <- summary(m1.aov)[[1]]
sigma2hat <- anova_table["Residuals", "Mean Sq"]

D <- 4 # Maximum mean difference to detect

# Computing effect size f
effect.size <- D / sqrt(sigma2hat)

pwr.anova.test(k=3, f=effect.size, power=0.99, sig.level = 0.05)  # Compute
required sample size
```

```
##
##      Balanced one-way analysis of variance power calculation
##
##               k = 3
##               n = 20.07823
##               f = 0.6121169
##       sig.level = 0.05
##           power = 0.99
##
## NOTE: n is number in each group
```

```r
# sample size n is 21, which means we need 21*3 = 63 total data points which
is less than what we have(90)
# for the effect size f = 0.6121169 our sample size is more than enough, our
sample size can sufficiently detect a difference of size 4(D) with a 99%
power.

#
==============================================================================
=====
# 3. RANDOMIZED BLOCK DESIGN: using misses the primary endpoint
#
==============================================================================
=====

# Response, treatment, and block respectively
misses = data$Misses
music = factor(data$Order, levels = c("N","S","L"))      # treatment label
participant = factor(data$Participants)                  # block label

RCBD_dataframe <- data.frame(misses,
                             factor(music),
                             factor(participant))

# (1) Statistical model
# y_{ij} = mu + tau_i + beta_j + e_{ij}
# for i = 1,2,3 and j = 1,...,30

# (2) Hypotheses for treatment effect
# H0: tau_1 = tau_2 = tau_3 = 0
# vs H1: tau_i is not 0 for at least one i

# (3) ANOVA table for RCBD
m1 = aov(misses ~ factor(music) + factor(participant), data = RCBD_dataframe)
summary(m1)
```

```
##                     Df Sum Sq Mean Sq F value Pr(>F)
## factor(music)        2  157.4   78.68   2.492 0.0916 .
## factor(participant) 29 1883.8   64.96   2.057 0.0098 **
## Residuals           58 1831.3   31.57
```

```
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

# (a) Decision for treatment effect
anova_table <- summary(m1)[[1]]
F0 <- anova_table["factor(music)", "F value"]
pvalue <- anova_table["factor(music)", "Pr(>F)"]
alpha <- 0.05

cat("Decision:", ifelse(pvalue < alpha, "Reject H0", "Do not reject H0"),
"\n")

## Decision: Do not reject H0

# to check whether blocking seem to be necessary,
MS_block <- anova_table["factor(participant)", "Mean Sq"]
MSE <- anova_table["Residuals", "Mean Sq"]
cat("MS(blocks) =", MS_block, "\n")

## MS(blocks) = 64.95824

cat("MSE =", MSE, "\n")

## MSE = 31.57433

# to check whether blocking seem to be necessary, we can Compare MS(blocks)
with MSE, form our results we get MS(blocks) = 64.95824 >>> MSE = 31.57433,
so yes blocking was indeed necessary.


# (4) we can also Compare the results (SSE) from RCBD with completely
randomized design
m2 = aov(misses ~ factor(music), data = RCBD_dataframe)
summary(m2)

##               Df Sum Sq Mean Sq F value Pr(>F)
## factor(music)  2    157   78.68   1.842  0.165
## Residuals     87   3715   42.70

# CRD SSE = 3715.1
# RCBD we have , SSE = 1831.3 and SSBlock = 1883.8
# eventhough we still reject the null hypothesis and say that music is not
significant, we noticed that the p value in RCBD (0.0916) is lower than the p
value in CRD (0.1646), which tells us that RCBD reduces the blocking effect
by reducing the variation between participants.

#
==========================================================================
=====
# 4. RANDOMIZED BLOCK DESIGN: using moves the secondary endpoint
#
```

```
==========================================================================
=====

moves = data$Moves
m1_moves = aov(moves ~ factor(music) + factor(participant), data =
RCBD_dataframe)
summary(m1_moves)

##                      Df Sum Sq Mean Sq F value Pr(>F)
## factor(music)         2  196.6   98.31   2.462 0.0941 .
## factor(participant) 29 2195.2   75.70   1.896 0.0193 *
## Residuals           58 2316.0   39.93
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

# (a) Decision for treatment effect
anova_table <- summary(m1_moves)[[1]]
F0 <- anova_table["factor(music)", "F value"]
pvalue <- anova_table["factor(music)", "Pr(>F)"]
alpha <- 0.05

cat("Decision:", ifelse(pvalue < alpha, "Reject H0", "Do not reject H0"),
"\n")

## Decision: Do not reject H0

# The Decision is still the same the type of music background has no
significance on memory performance using either endpoints

#
==========================================================================
=====
# 5.FACTORIAL DESIGN: 3 Factor factorial design (factors: music, age, gender)
#
==========================================================================
=====
# required data
music  <- factor(data$Order, levels = c("N","S","L"))

gender <- factor(data$Gender,
                 levels = c("F","M"),
                 labels = c("Female","Male"))

age <- factor(data$Age,
              levels = c("Y","O"),
              labels = c("Young","Old"))

# Response
response <- data$Misses
```

```r
# Three-factor factorial design

model1 <- aov(response ~ music * gender * age)
summary(model1)
```

```
##                   Df Sum Sq Mean Sq F value Pr(>F)
## music              2    157   78.68   1.856 0.1631
## gender             1      1    1.34   0.032 0.8591
## age                1    243  242.81   5.728 0.0191 *
## music:gender       2     83   41.74   0.985 0.3781
## music:age          2     45   22.29   0.526 0.5932
## gender:age         1      8    8.14   0.192 0.6625
## music:gender:age   2     28   14.03   0.331 0.7193
## Residuals         78   3307   42.39
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
# The 3 factor interaction, music and gender are not significant since the p
values for each are greater than 0.05, but age is significant since our p
value = 0.0191 is less than 0.05.


#
==========================================================================
=====
# 6.FACTORIAL DESIGN + RBCD: 3 Factor factorial design (factors: music, age,
gender) and RBCD, taking
#   participants as a block to reduce between partcipant variation
#
==========================================================================
=====


# to reduce the variation between participant to partipant, we have coupled
factorial design and RBCD
model2 <- aov(Misses ~ music * gender * age + factor(Participants), data =
data)
summary(model2)
```

```
##                       Df Sum Sq Mean Sq F value  Pr(>F)
## music                  2  157.4   78.68   2.442 0.09688 .
## gender                 1    1.3    1.34   0.042 0.83893
## age                    1  242.8  242.81   7.537 0.00828 **
## factor(Participants) 27 1639.6   60.73   1.885 0.02489 *
## music:gender           2   83.5   41.74   1.296 0.28237
## music:age              2   44.6   22.29   0.692 0.50519
## music:gender:age       2   28.1   14.03   0.435 0.64935
```

```
## Residuals            52 1675.2   32.22
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
# The 3 factor interaction, music and gender are not significant since the p
values for each are greater than 0.05, but age is still significant since our
p value = 0.00828 is less than 0.05.

# Since our results from the factorial design suggested that age is a
significant factor we can test which age group is performing well in the
memory puzzel game
TukeyHSD(aov(Misses ~ age, data = data))
```

```
##   Tukey multiple comparisons of means
##     95% family-wise confidence level
##
## Fit: aov(formula = Misses ~ age, data = data)
##
## $age
##               diff      lwr      upr     p adj
## Old-Young 3.417863 0.626613 6.209113 0.0169792
```

```r
# from this result we get that participants that are old has 3.417863 more
misses than the young participants on   average.

#
==========================================================================
=====
# 7.FACTORIAL DESIGN,FACTORIAL DESIGN + RBCD: using moves the secondary
endpoint
#
==========================================================================
=====
response <- data$Moves

# Factorial design
model_moves <- aov(response ~ music * gender * age)
summary(model_moves)
```

```
##                  Df Sum Sq Mean Sq F value Pr(>F)
## music             2    197    98.3   1.952 0.1489
## gender            1     11    11.4   0.226 0.6359
## age               1    323   322.7   6.408 0.0134 *
## music:gender      2     86    42.8   0.851 0.4311
## music:age         2     35    17.6   0.349 0.7066
## gender:age        1     43    43.0   0.854 0.3583
## music:gender:age  2     85    42.3   0.841 0.4353
## Residuals        78   3929    50.4
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
#factorial design + RCBD
model_moves1 <- aov(response ~ music * gender * age + factor(Participants),
data = data)
summary(model_moves1)
```

```
##                      Df Sum Sq Mean Sq F value  Pr(>F)
## music                 2  196.6    98.3   2.422 0.09867 .
## gender                1   11.4    11.4   0.280 0.59874
## age                   1  322.7   322.7   7.952 0.00678 **
## factor(Participants) 27 1861.0    68.9   1.698 0.05053 .
## music:gender          2   85.7    42.8   1.056 0.35532
## music:age             2   35.1    17.6   0.433 0.65094
## music:gender:age      2   84.7    42.3   1.043 0.35958
## Residuals            52 2110.5    40.6
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
# even taking the secondary endpoint, moves suggests that only age is
significant
```